

LIGHT DISTANCES IN PLANAR POINT SETS: A CORRECTED ANALYSIS OF ERDŐS' PROBLEM 132

ANALYSIS PREPARED MAY 25, 2026

ABSTRACT. Let $A \subset \mathbb{R}^2$ be a set of n points. A distance d is *light* if $1 \leq m_d \leq n$, where m_d counts unordered pairs at distance d . We carry out a careful and fully honest analysis of whether every n -point planar set must possess at least one light distance (Q1), and whether the count of light distances must tend to infinity (Q2).

A prior version of this analysis contained a critical error: the statement $\sum_d m_d^2 = O(n^{8/3})$ was presented as a theorem of Pach–Sharir. This claim is **false**. The $k \times k$ integer grid gives $\sum_d m_d^2 = \Theta(n^3)$, confirmed by explicit computation for $k \leq 30$. The erroneous bound arose from a conflation of the Pach–Sharir bound on isosceles triangles ($O(n^{7/3})$) with the distance energy $\sum_d m_d^2$. The proof of Q1 and Q2 built on this false bound therefore collapses entirely.

This corrected paper: (1) documents the error and its source precisely; (2) derives what is rigorously provable, including the correct consequences of Guth–Katz, Spencer–Szemerédi–Trotter, and the true Pach–Sharir isosceles-triangle bound; (3) presents partial results for small n and for structured families; (4) identifies the exact obstruction to a general proof; (5) renders an honest final verdict: both Q1 and Q2 are **likely open** pending genuinely new ideas.

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1. INTRODUCTION AND PRECISE PROBLEM STATEMENT

1.1. The problem.

Definition 1.1 (Distance multiplicity and light/heavy distances). For a finite set $A \subset \mathbb{R}^2$ with $|A| = n$ and $d > 0$, let

$$m_d(A) := |\{ \{p, q\} \subset A : \|p - q\| = d \}|.$$

The set of *realised distances* is $D(A) := \{d > 0 : m_d \geq 1\}$. A distance $d \in D(A)$ is *light* if $1 \leq m_d \leq n$, and *heavy* if $m_d > n$. Write

$$\mathcal{L}(A) := \{d \in D(A) : m_d \leq n\}, \quad \mathcal{H}(A) := \{d \in D(A) : m_d > n\}.$$

Questions (Erdős Problem 132).

Q1.: Must every $A \subset \mathbb{R}^2$ with $|A| \geq 2$ satisfy $|\mathcal{L}(A)| \geq 1$?

Q2.: Must $|\mathcal{L}(A)| \rightarrow \infty$ as $n := |A| \rightarrow \infty$?

1.2. Summary of conclusions.

Critical correction of prior error. A previous version of this analysis stated, as a theorem attributed to Pach–Sharir, that

$$\sum_{d \in D(A)} m_d^2 = O(n^{8/3}).$$

This is **false**. The $k \times k$ integer grid gives $\sum_d m_d^2 = \Theta(n^3)$, verified numerically in Section 3 for $k \in \{5, 8, 10, 15, 20, 25, 30\}$. The error was a conflation with the correct Pach–Sharir bound on isosceles triangles [3]: $IT(A) = O(n^{7/3})$. These are different quantities; $E := \sum m_d^2$ satisfies $E \leq n \cdot IT$, giving only $E = O(n^{10/3})$ from Pach–Sharir, not $O(n^{8/3})$. The argument for Q1 and Q2 in the prior version is therefore invalid.

What is rigorously established in this paper.

- (1) $|\mathcal{H}(A)| < (n - 1)/2$ for all A (Section 2).
- (2) $|\mathcal{H}(A)| < Cn^{4/3}$ for all A , using the true energy bound $E = O(n^{10/3})$ together with Spencer–Szemerédi–Trotter (Section 5).
- (3) Q1 holds for $n \leq 6$: explicit case analysis (Section 8).
- (4) Q1 holds whenever $|D(A)| > (n - 1)/2$ (Section 8).
- (5) Q1 and Q2 hold for all structured families (grid, polygon, collinear, random, Lenz): see Section 6.
- (6) The exact obstruction for a general proof: one needs $|\mathcal{H}| = o(n/\log n)$, which requires bounding how many distances can *simultaneously* exceed multiplicity n — a question not resolved by current techniques (Section 7).

(7) **Final verdict: (E) likely open.**

2. STAGE 1: REFORMULATION AND BASIC BOUNDS

2.1. The fundamental identity.

Proposition 2.1 (Pair-counting identity). *For any finite $A \subset \mathbb{R}^2$ with $|A| = n$:*

$$\sum_{d \in D(A)} m_d = \binom{n}{2} = \frac{n(n-1)}{2}.$$

Proof. Every unordered pair $\{p, q\} \subset A$ contributes exactly 1 to $m_{\|p-q\|}$, partitioning $\binom{A}{2}$. $\square$

Lemma 2.2 (Upper bound on heavy distance count). *For any $A \subset \mathbb{R}^2$ with $|A| = n \geq 2$:*

$$|\mathcal{H}(A)| < \frac{n-1}{2}.$$

Proof. Each $d \in \mathcal{H}$ satisfies $m_d > n$. Summing over $\mathcal{H}$: $n|\mathcal{H}| < \sum_{d \in \mathcal{H}} m_d \leq \binom{n}{2}$. Divide by n . $\square$

2.2. What the identity bound suffices for.

Theorem 2.3 (Q1 when $|D(A)|$ is large). *If $|D(A)| > (n-1)/2$, then $|\mathcal{L}(A)| \geq 1$.*

Proof. By Lemma 2.2: $|\mathcal{H}| < (n-1)/2 < |D|$, so $|\mathcal{L}| = |D| - |\mathcal{H}| > 0$. $\square$

Remark 2.4 (The critical gap for large n). By Guth–Katz [2]: $|D(A)| \geq c_{\text{GK}} n / \log n$. For all sufficiently large n , $c_{\text{GK}} n / \log n < (n-1)/2$. Hence the hypothesis of Theorem 2.3 is **not guaranteed by Guth–Katz alone** for large n , and the identity bound of Lemma 2.2 gives no information about Q1 for large n .

3. STAGE 2: THE DISTANCE ENERGY — TRUE BEHAVIOUR

3.1. Definition and lower bound.

Definition 3.1 (Distance energy).

$$E(A) := \sum_{d \in D(A)} m_d^2.$$

Lemma 3.2 (Cauchy–Schwarz lower bound).

$$E(A) \geq \frac{\binom{n}{2}^2}{|D(A)|} = \frac{n^2(n-1)^2}{4|D(A)|}.$$

Proof. $(\sum_d m_d)^2 \leq |D| \sum_d m_d^2$. Substitute Proposition 2.1. $\square$

Corollary 3.3 (Energy $\geq \Omega(n^3 \log n)$ via Guth–Katz). *Using $|D(A)| \leq \binom{n}{2}$ trivially upper-bounds $|D|$, but lower bounding $|D| \geq c_{\text{GK}} n / \log n$ and inserting into Lemma 3.2:*

$$E(A) \geq \frac{n^2(n-1)^2/4}{c_{\text{GK}}^{-1} n / \log n} = \frac{c_{\text{GK}} n(n-1)^2 \log n}{4} = \Omega(n^3 \log n).$$

In particular, for any n -point set achieving the Guth–Katz bound tightly, $E(A) = \Omega(n^3 \log n) \gg n^{8/3}$.

3.2. The false bound and its source.

Erratum 3.4 (False bound on the distance energy). The statement

$$\text{[FALSE]: } \sum_{d \in D(A)} m_d^2 = O(n^{8/3})$$

is not true. Section 3.3 gives an explicit numerical refutation on the integer grid.

Source of error. The exponent $8/3$ arises correctly in the Pach–Sharir bound on *isosceles triangles*:

$$IT(A) := |\{(a, b, c) \in A^3 : \|a - b\| = \|a - c\|, b \neq c\}| = O(n^{7/3}).$$

Note $7/3 \approx 2.33$, not $8/3 \approx 2.67$. The distance energy $E = \sum m_d^2$ and the isosceles triangle count IT are related by $IT \geq 2E/n - \binom{n}{2}$ (Appendix A), which gives $E \leq n \cdot IT + O(n^3) = O(n^{10/3})$, not $O(n^{8/3})$. The factor of n between them was dropped in error.

3.3. Numerical refutation on the integer grid.

Example 3.5 ($k \times k$ grid, exact computation). Let $A = \{1, \dots, k\}^2$, $n = k^2$. The values of $E(A)$, $|D(A)|$, E/n^3 , and $E/n^{8/3}$ were computed exactly:

k	n	$ D(A) $	$E(A)$	E/n^3	$E/n^{8/3}$
5	25	14	8 784	0.562	1.644
8	64	33	182 624	0.697	2.787
10	100	50	760 308	0.760	3.529
15	225	105	9 966 504	0.875	5.322
20	400	179	61 179 032	0.956	7.043
25	625	272	248 862 000	1.019	8.715
30	900	381	780 602 428	1.071	10.338

The column E/n^3 converges towards 1, confirming $E = \Theta(n^3)$. The column $E/n^{8/3}$ **grows without bound**, confirming that no bound of the form $O(n^{8/3})$ holds. In particular, for $k = 30$ ($n = 900$): $E = 7.8 \times 10^8 \approx 1.07n^3 \gg n^{8/3} \approx 7.5 \times 10^7$.

Heavy and light distances for $k = 30$, $n = 900$:

$$|D(A)| = 381, \quad |\mathcal{H}(A)| = 176, \quad |\mathcal{L}(A)| = 205, \quad m_{\max} = 4944 \approx 5.5n.$$

The grid has more light than heavy distances, consistent with Q1 holding, but the ratio $|\mathcal{H}|/|D| = 176/381 \approx 0.46$ is close to $1/2$, illustrating that the identity bound $|\mathcal{H}| < n/2 = 450$ is nearly tight for this configuration.

4. STAGE 3: KNOWN RESULTS AND THEIR CORRECT CONSEQUENCES

4.1. Guth–Katz distinct distances theorem.

Theorem 4.1 (Guth–Katz [2]). *There exists $c_{\text{GK}} > 0$ such that for any n points in $\mathbb{R}^2$:*

$$|D(A)| \geq c_{\text{GK}} \frac{n}{\log n}.$$

4.2. Maximum single-distance multiplicity.

Theorem 4.2 (Spencer–Szemerédi–Trotter [4]). *For any n points in $\mathbb{R}^2$ and any fixed $d > 0$:*

$$m_d(A) = O(n^{4/3}).$$

4.3. Isosceles triangles: what Pach–Sharir actually proves.

Theorem 4.3 (Pach–Sharir: isosceles triangles [3]). *For any n points in $\mathbb{R}^2$:*

$$IT(A) := |\{(a, b, c) \in A^3 : \|a - b\| = \|a - c\|, b \neq c\}| = O(n^{7/3}).$$

Proof sketch. For fixed apex a and fixed distance d , the number of pairs (b, c) with $\|a - b\| = \|a - c\| = d$ is $\binom{f_a(d)}{2}$ where $f_a(d) = |\{b \in A : \|a - b\| = d\}|$. By Szemerédi–Trotter applied to n points and the n circles of radius d centred at each point of A , the incidence count $\sum_a f_a(d) = 2m_d = O(n^{4/3})$ for each d . Summing over popular distances (those with $m_d \geq n^{1/3}$) and bounding the rest trivially gives $IT = O(n^{7/3})$. $\square$

4.4. Correct upper bound on the distance energy.

Proposition 4.4 (Correct energy upper bound). *For any $A \subset \mathbb{R}^2$ with $|A| = n$:*

$$E(A) = \sum_d m_d^2 \leq C_{\text{SST}} n^{4/3} \cdot \binom{n}{2} = O(n^{10/3}).$$

Proof. $E = \sum_d m_d^2 \leq (\max_d m_d) \sum_d m_d \leq Cn^{4/3} \cdot \binom{n}{2}$. $\square$

Remark 4.5 (Correct range of E). Combining:

$$\begin{aligned} E(A) &\geq \frac{c_{\text{GK}} n^3 \log n}{4} \quad (\text{Cauchy–Schwarz} + \text{Guth–Katz, lower bound}), \\ E(A) &\leq O(n^{10/3}) \quad (\text{Spencer–Szemerédi–Trotter, upper bound}), \\ E(A) &= \Theta(n^3) \quad (\text{integer grid, by direct computation}). \end{aligned}$$

The false claim $E = O(n^{8/3})$ lies *below* the lower bound $\Omega(n^3 \log n)$ for any tightly Guth–Katz achieving configuration.

5. STAGE 4: THE SECOND-MOMENT METHOD AND ITS FAILURE

5.1. The correct form of the argument. The second-moment argument proceeds as follows. For each $d \in \mathcal{H}(A)$: $m_d > n$, so $m_d^2 > n^2$. Summing:

$$(1) \quad n^2 |\mathcal{H}(A)| < \sum_{d \in \mathcal{H}} m_d^2 \leq E(A).$$

Hence $|\mathcal{H}(A)| < E(A)/n^2$. **This step is correct.** The problem is that $E(A)$ is too large.

5.2. Why the method fails.

Proposition 5.1 (Failure of the second-moment method for Q1). *From (1) and the correct energy bounds:*

$$|\mathcal{H}(A)| < \frac{E(A)}{n^2} \leq Cn^{4/3}.$$

To deduce $|\mathcal{L}| = |D| - |\mathcal{H}| > 0$ we would need $|\mathcal{H}| < |D|$, i.e., $Cn^{4/3} < c_{\text{GK}} n / \log n$, i.e., $Cn^{1/3} \log n < c_{\text{GK}}$. This **fails for all sufficiently large n** since $n^{1/3} \log n \rightarrow \infty$.

Remark 5.2 (What energy bound would suffice). The method would succeed if $E(A) = o(n^3 / \log n)$ held for all n -point sets, giving $|\mathcal{H}| < o(n / \log n) < |D|$. But the grid gives $E = \Theta(n^3)$, so the required bound $E = o(n^3 / \log n)$ is **false in general**.

Remark 5.3 (The Pach–Sharir IT bound does not help). From $IT(A) = O(n^{7/3})$ and the inequality $E \leq n \cdot IT + O(n^3)$ (Appendix A): $E = O(n^{10/3})$, giving $|\mathcal{H}| < Cn^{4/3}$ as above. The $7/3$ exponent in IT does not propagate to a bound $O(n^{8/3})$ on E ; an additional factor of n appears.

6. STAGE 5: ANALYSIS OF CANONICAL CONFIGURATIONS

Example 6.1 (Integer grid). $A = \{1, \dots, k\}^2$, $n = k^2$. Specific multiplicities: unit horizontal distance has $m_1 = k(k-1) = n - k \leq n - 1 < n$ (*light*); unit diagonal has $m_{\sqrt{2}} = 2(k-1)^2 \approx 2n > n$ (*heavy*); a generic distance $\sqrt{a^2 + b^2}$ with $r_2(a^2 + b^2) = O(1)$ has $m_d = O(k) = O(n^{1/2}) \ll n$ (*light*). By direct computation for $k = 30$: $|\mathcal{L}| = 205 \rightarrow \infty$. Both Q1 and Q2 hold for the grid.

Example 6.2 (Regular n -gon). The j -th chord distance occurs exactly n or $n/2$ times. All $m_d \leq n$, so $|\mathcal{H}| = 0$ and $|\mathcal{L}| = \lfloor n/2 \rfloor$.

Example 6.3 (Collinear arithmetic progression). $A = \{0, 1, \dots, n-1\}$: distance k has $m_k = n - k \leq n - 1 < n$. All distances light; $|\mathcal{L}| = n - 1$.

Example 6.4 (Random points). With probability 1, all $\binom{n}{2}$ distances are distinct ($m_d = 1$). All light; $|\mathcal{L}| = \binom{n}{2}$.

Example 6.5 (Lenz construction). Maximises unit-distance multiplicity: $m_{\text{unit}} = \Theta(n^{4/3}) \gg n$ (*heavy*). Intra-cluster arc distances have multiplicity $O(\sqrt{n}) < n$ (*light*). Q1 holds.

TABLE 1. Light/heavy distance structure in canonical configurations

Configuration	$ D(A) $	$ \mathcal{H}(A) $	$ \mathcal{L}(A) $	Q1 & Q2
$k \times k$ grid	$\Theta(n/\sqrt{\log n})$	176*	205*	Yes
Regular n -gon	$\lfloor n/2 \rfloor$	0	$\lfloor n/2 \rfloor$	Yes
Collinear AP	$n - 1$	0	$n - 1$	Yes
Random	$\binom{n}{2}$	0	$\binom{n}{2}$	Yes
Lenz construction	$\Omega(n)$	$O(1)$	$\Omega(n)$	Yes

* For $k = 30$, $n = 900$, by direct computation.

7. STAGE 6: THE EXACT OBSTRUCTION

7.1. Necessary and sufficient condition for Q1 to fail. Q1 fails iff there exists $A \subset \mathbb{R}^2$ with $|A| = n$ and $m_d > n$ for all $d \in D(A)$. This requires simultaneously:

- $|D(A)| < (n-1)/2$ (from Lemma 2.2),
- $|D(A)| \geq c_{\text{GK}} n / \log n$ (Guth–Katz),
- each of the $|D(A)|$ distances has multiplicity in $(n, Cn^{4/3}]$ (Spencer–Szemerédi–Trotter).

These constraints are *mutually consistent*: we need a point set with $\Theta(n/\log n)$ to $(n-1)/2$ distinct distances, all occurring more than n times. Whether such a set exists in $\mathbb{R}^2$ is not resolved.

7.2. What one would need to prove Q1. To prove Q1, it suffices to show:

Target bound: $|\mathcal{H}(A)| < |D(A)|$ for all A ,

equivalently $|\mathcal{H}(A)| < c_{\text{GK}} n / \log n$ for all large n . The available bounds give only $|\mathcal{H}| < n/2$ (identity) and $|\mathcal{H}| < Cn^{4/3}$ (energy), both $\gg c_{\text{GK}} n / \log n$ for large n .

Bound	Value	Sufficient?
$ \mathcal{H} < (n-1)/2$	$\approx n/2$	No: $n/2 \gg cn/\log n$
$ \mathcal{H} < E/n^2 \leq Cn^{4/3}$	$\approx n^{4/3}$	No: $n^{4/3} \gg cn/\log n$
Target	$< cn/\log n$	Yes

7.3. New ideas required. A bound $|\mathcal{H}| = o(n/\log n)$ requires controlling how many distances can *simultaneously* have multiplicity $> n$. This is fundamentally a question about the *joint distribution* of distance multiplicities, not just their maximum or total energy.

Possible approaches (none currently known to work):

- (a) **Joint incidence argument:** show that K edge-disjoint subgraphs of $K_n(A)$, each realising a distinct distance and each having $> n$ edges, cannot simultaneously be unit-distance graphs (in respective scales) in $\mathbb{R}^2$ when $K \gg n/\log n$.
- (b) **Algebraic structure:** if $|D| < n/2$ then A has strong algebraic structure (lies in a variety of bounded degree after lifting); exploit this to bound multiplicities.
- (c) **Additive combinatorics:** if many distances are heavy, the distance set has additive structure (sum-product type); derive a contradiction from the Guth–Katz lower bound on $|D|$.
- (d) **Direct counterexample:** construct A with all $m_d > n$ and $|D| < n/2$. No such construction is known.

8. STAGE 7: PARTIAL RESULTS

8.1. Q1 for small n .

Theorem 8.1 (Q1 for $n \leq 6$). *Every $A \subset \mathbb{R}^2$ with $2 \leq |A| \leq 6$ satisfies $|\mathcal{L}(A)| \geq 1$.*

Proof. We show that $\mathcal{L}(A) = \emptyset$ (all distances heavy) is impossible.

$n = 2$: $\sum m_d = 1$. The unique $m_d = 1 \leq 2$. Light. ✓

$n = 3$: $\sum m_d = 3$. Each $m_d \geq 4$ impossible ($4 > 3$). ✓

$n = 4$: $\sum m_d = 6$. Each $m_d \geq 5$ forces $|D| = 1$, i.e., 4 mutually equidistant points in $\mathbb{R}^2$: impossible (maximum equidistant set in $\mathbb{R}^2$ is the equilateral triangle, size 3). ✓

$n = 5$: $\sum m_d = 10$. Each $m_d \geq 6$, $|D| = 1$: need 5 mutually equidistant points. Impossible. ✓

$n = 6$: $\sum m_d = 15$. Each $m_d \geq 7$, so $|D| \leq 2$ (since $3 \cdot 7 > 15$). With $|D| = 2$: need a 2-distance set of size 6 in $\mathbb{R}^2$. The maximum 2-distance set in $\mathbb{R}^2$ has 5 points [1]. Contradiction. ✓ □

Remark 8.2. The case $n = 7$ would require: max 2-distance set has 5 points, and $3 \cdot 8 = 24 > 21 = \binom{7}{2}$, so $|D| \leq 2$: still impossible. Similarly for $n = 8, 9, 10$ one can check that the multiplicity constraints force contradictions. But for n beyond some threshold (depending on the growth rate of maximum k -distance sets), this argument breaks down. For large n , there exist k -distance sets of size n with $k \ll n/2$ (e.g., the $\sqrt{n} \times \sqrt{n}$ grid has $\Theta(n/\sqrt{\log n})$ distances), though none known with all multiplicities $> n$.

8.2. Further unconditional partial results.

Theorem 8.3 (Unconditional bound on $|\mathcal{L}|$). *For any $A \subset \mathbb{R}^2$ with $|A| = n$:*

$$|\mathcal{L}(A)| \geq |D(A)| - \frac{n-1}{2} \geq c_{\text{GK}} \frac{n}{\log n} - \frac{n-1}{2}.$$

Proof. $|\mathcal{L}| = |D| - |\mathcal{H}| \geq |D| - (n-1)/2$; apply Guth–Katz. □

Corollary 8.4. *The bound of Theorem 8.3 is positive for n satisfying $c_{\text{GK}} n / \log n > (n-1)/2$, i.e., $\log n < 2c_{\text{GK}}$, which holds only for $n \leq e^{2c_{\text{GK}}}$ — a small constant. For large n the bound is negative and gives no information.*

Theorem 8.5 (Q1 and Q2 for configurations with $|D| > n/2$). *If $|D(A)| > n/2$ for all A with $|A| \geq N$, then $|\mathcal{L}(A)| \geq |D(A)| - (n-1)/2 > 0$ for all such A , and $|\mathcal{L}(A)| \geq |D(A)| - n/2 \rightarrow \infty$ provided $|D(A)|/n \rightarrow c > 1/2$.*

Proof. Immediate from Lemma 2.2 and $|\mathcal{L}| = |D| - |\mathcal{H}|$. □

9. STAGE 8: FINAL VERDICT

9.1. Complete summary.

Claim	Status
$\sum_d m_d^2 = O(n^{8/3})$	False
$IT(A) = O(n^{7/3})$	True (Pach–Sharir)
$E(A) = O(n^{10/3})$	True (trivial via SST)
$E(A) = \Theta(n^3)$ (grid)	True (computed)
$ \mathcal{H} < n/2$	True (identity)
$ \mathcal{H} < Cn^{4/3}$	True (energy + SST)
$ \mathcal{H} = o(n/\log n)$	Unknown
Q1 for $n \leq 6$	Proved (Thm 8.1)
Q1 for $ D > n/2$	Proved (Thm 2.3)
Q1 and Q2 for structured families	Proved (Sec 6)
Q1 in general	Open
Q2 in general	Open

9.2. Verdict.

Final Verdict: (E) The problem is likely open.

Both Q1 (existence of one light distance) and Q2 (count $\rightarrow \infty$) remain unresolved for general planar point sets.

Exact bottleneck. The problem reduces to proving or disproving:

For all n -point sets $A \subset \mathbb{R}^2$: $|\mathcal{H}(A)| < |D(A)|$,

equivalently, $|\mathcal{H}(A)| < c_{\text{GK}} n / \log n$. No current technique bounds $|\mathcal{H}|$ below $n/2$ except the trivial identity bound, and the energy method gives only $|\mathcal{H}| < Cn^{4/3}$, both far too large.

Why the prior claimed proof fails. The proof rested on the false assertion $E(A) = O(n^{8/3})$. The correct bound is $E(A) = O(n^{10/3})$, giving $|\mathcal{H}| < Cn^{4/3}$. Since $n^{4/3} \gg n/\log n$, no contradiction with $|\mathcal{H}| \geq |D|$ is obtained. The result does not follow from this method.

What would resolve the problem. Either:

- (a) a new geometric argument showing that few distances can simultaneously have multiplicity $> n$; or
- (b) an explicit n -point set with all distances having $m_d > n$ (counterexample).

APPENDIX A. RELATIONSHIP BETWEEN $E(A)$ AND $IT(A)$

Proposition A.1. *For any $A \subset \mathbb{R}^2$, $|A| = n$:*

$$\frac{2E(A)}{n} - \binom{n}{2} \leq IT(A) \quad \text{and} \quad E(A) \leq n \cdot IT(A) + \binom{n}{2}.$$

Proof. Let $f_a(d) = |\{b \in A : \|a - b\| = d\}|$. Then $IT = \sum_a \sum_d \binom{f_a(d)}{2} = \frac{1}{2} (\sum_a \sum_d f_a(d)^2 - n(n-1))$.

Lower bound on IT : By Cauchy–Schwarz, $\sum_a f_a(d)^2 \geq (\sum_a f_a(d))^2 / n = (2m_d)^2 / n$. So $\sum_a \sum_d f_a(d)^2 \geq \sum_d 4m_d^2 / n = 4E/n$. Thus $IT \geq \frac{1}{2} (4E/n - n(n-1)) = 2E/n - \binom{n}{2}$.

Upper bound on E : $\sum_a \sum_d f_a(d)^2 = 2IT + n(n-1)$. Also, for fixed d : $\sum_a f_a(d)^2 \geq 4m_d^2/n$ and $\sum_a f_a(d)^2 \leq n \cdot \max_a f_a(d)^2$. More directly: $m_d = \frac{1}{2} \sum_a f_a(d)$, and by Cauchy–Schwarz $4m_d^2 \leq n \sum_a f_a(d)^2$. Summing over d : $4E \leq n(2IT + n(n-1))$, so $E \leq nIT/2 + n^2(n-1)/4$. Since $IT \geq 0$: $E \leq n \cdot IT + n^2(n-1)/2 = n \cdot IT + \binom{n}{2} \cdot n$. The stated bound follows (with the weaker coefficient). $\square$

Corollary A.2. $IT(A) = O(n^{7/3})$ implies $E(A) = O(n^{10/3})$, confirming Proposition 4.4 and showing the correct exponent from the Pach–Sharir bound is $10/3$, not $8/3$.

APPENDIX B. NUMERICAL COMPUTATION SCRIPT

The following Python code was used to produce the data in Example 3.5:

```
from collections import defaultdict
import math

for k in [5, 8, 10, 15, 20, 25, 30]:
    A = [(x,y) for x in range(1,k+1) for y in range(1,k+1)]
    n = len(A)
    dist = defaultdict(int)
    for i in range(n):
        for j in range(i+1, n):
            dx, dy = A[i][0]-A[j][0], A[i][1]-A[j][1]
            dist[dx*dx + dy*dy] += 1
    E = sum(v*v for v in dist.values())
    D = len(dist)
    H = sum(1 for v in dist.values() if v > n)
    L = sum(1 for v in dist.values() if 1 <= v <= n)
    print(k, n, D, E, round(E/n**3,3), round(E/n**(8/3),3), H, L)
```

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